Rodrigo G. Freundt

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Summary

- Research-oriented engineer and PhD candidate in Astronomy and Space Sciences with +10 years of experience building and integrating astronomical instrumentation. Excellent technical writing, project management and mentoring skills. Extensive experience working in large cross-functional collaborations with diverse teams of scientists and engineers.

Education

Cornelll University Ithaca, NY
Ph.D. in Astronomy and Space Sciences. 2021–Present

- Thesis: "The Epoch of Reionization Spectrometer (EoR-Spec): Development and Early Science." (in preparation). Expected completion: June 2027.

Pontifical Catholic University of Peru Lima, Peru
B.Sc. in Electronics Engineering. 2011–2016

- Thesis: "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU".

Experience

CCAT collaboration Ithaca, NY.
Graduate Research Assistant. Advisors: Prof. Gordon Stacey and Prof. Michael Niemack 2021 –Present

- Leading the design, fabrication, testing and eventual deployment of EoR-Spec, a direct-detection imaging spectrometer designed to obtain tomographic maps of the early universe via Line Intensity Mapping (LIM) of redshifted [C II] 158 μm fine-structure line emission ($z \sim 3.5 - 8$).
- Designed and integrated a cryogenic readout harness for the Prime-Cam receiver, a modular cryogenic camera for the Fred Young Submillimeter Telescope (FYST).

Atacama Cosmology Telescope San Pedro de Atacama, Chile.
Site Engineer. Advisors: Prof. Suzanne Staggs and Prof. Mark Devlin. Oct 2019 –Jul 2021

- Ensured the continuous operation of complex astronomical instrumentation at extreme altitude (17,000 ft). In charge of regular maintenance, troubleshooting and overhaul of electrical, cryogenic and mechanical systems.
- Performed calibration procedures, such as the alignment of the receiver with the primary and secondary reflectors through high-precision photogrammetry ($\sim$10μm). Carried out seasonal passband characterization via Fourier Transform Spectroscopy.
- Helped in the deployment of the last detector array upgrade (27/39 GHz) for the ACTPol receiver.

Management Solutions Lima, Peru / Madrid, Spain.
Assistant Business Consultant Sep 2018 –Jul 2019

- Worked in big data and bank stress test for a major local bank in Lima, Peru. Software development and data visualization with interactive dashboards for a leading financial institution in Madrid, Spain.

Institute for Radioastronomy Lima, Peru.
Research Engineer (Intern until Apr 2017) Advisor: Prof. Jorge Heraud Mar 2015 –Aug 2018

- Worked in the construction of an 8-meter radio telescope (RT8) intended for teaching and outreach. Designed hardware and software for the telescope's electromechanical pointing system.

- Developed custom firmware for the backend electronics of the telescope.
- Assisted in the characterization of a 1.42 GHz feed antenna using software-defined radio technology.

Skills

- **Programming:** Python, C/C++, CUDA, LabVIEW, microcontroller firmware.
- **CAD/Simulation** Solidworks, KiCAD, Ansys Zemax Optics Studio.
- **Tools:** LaTeX, Git, Docker, Linux.

Awards

- Beca de Estímulo Académico Solidario (BEAS) 2013–2016
  Fully-funded undergraduate scholarship for outstanding academic performance and financial need.
- URSI Young Scientist Award Aug 2017
  Merit-based registration fee waiver and travel grant to attend URSI 2017, Montreal, QC.
- Summer School of Astronomical Instrumentation, Dunlap Institute, UofT Jul 2019
  Merit-based tuition waiver and travel grant. Toronto, ON.
- CASPER Workshop, Harvard & Smithsonian Center for Astrophysics (CfA) Aug 2019
  Merit-based fee waiver and travel grant (invited, could not attend). Cambridge, MA.

Teaching

- **Teaching Assistant** at Cornell University. Fall 2022, 2024, 2025, 2026
  Multiwavelenght astronomical techniques (ASTRO 4410)
  Advisors: Prof. Jim Cordes and Prof. Gordon Stacey
- **Teaching Assistant** at Cornell University Spring 2023
  Our Solar System (ASTRO 1102)
- **Teaching Assistant** at PUCP Aug 2017 - Dec 2017
  Space Science and Engineering (ING 306)

First-Author Publications

[1] **R. Freundt**, Y. Li, D. Henke, J. Austermann, J. R. Burgoyne, S. Chapman, S. K. Choi, C. J. Duell, Z. Huber, M. Niemack, T. Nikola, L. Lin, D. A. Riechers, G. Stacey, A. K. Vaskuri, E. M. Vavagiakis, J. Wheeler, and B. Zou, "CCAT: A status update on the EoR-Spec instrument module for Prime-Cam", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 343–352.

[2] **R. Freundt** and J. A. Heraud, "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU", in *2017 XXXIInd General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*, Aug. 2017, pp. 1–4.

Collaboration Publications

[3] Lin, L. T. et al. (including **R. G. Freundt**), "Ccat: Mod-cam cryogenic performance and its impact on 280-ghz kid array noise", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–7, 2026, ISSN: 2378-7074.

[4] Patel, D. A. et al. (including **R. G. Freundt**), "Ccat: Design, implementation, and testing of a system to read out over 10 000 280 ghz kids using rfsoc electronics", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 2 503 607–2 503 607, 2026, ISSN: 2378-7074.

[5] Calabrese, E. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 constraints on extended cosmological models", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 063, Nov. 2025, ISSN: 1475-7516.

[6] Louis, T. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 power spectra, likelihoods and lcdm parameters", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 062, Nov. 2025, ISSN: 1475-7516.

[7] Middleton, A. et al. (including **R. G. Freundt**), "Ccat: Led mapping and characterization of the 280 ghz tin kid array", *IEEE Transactions on Applied Superconductivity*, vol. 35, no. 5, pp. 1–4, 2025.

[8] Naess, S. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 maps", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 061, Nov. 2025, ISSN: 1475-7516.

[9] Vaskuri, A. K. et al. (including **R. G. Freundt**), "280-GHz aluminum MKID arrays for the Fred Young Submillimeter Telescope", *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 11, 026005, p. 026 005, Apr. 2025.

[10] Coulton, W. et al. (including **R. G. Freundt**), "Atacama cosmology telescope: High-resolution component-separated maps across one third of the sky", *Physical Review D*, vol. 109, no. 6, Mar. 2024, ISSN: 2470-0029.

[11] B. Zou, J. R. Bond, V. Butler, **R. Freundt**, Z. B. Huber, M. D. Niemack, T. Nikola, G. J. Stacey, and E. M. Vavagiakis, "CCAT: Characterizing the Fabry-Pérot interferometer mirrors and mount for the epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 909–916.

[12] Choi, S. K. et al. (including **R. G. Freundt**), "Ccat-prime: Characterization of the first 280 ghz mkid array for prime-cam", *Journal of Low Temperature Physics*, vol. 209, no. 5-6, pp. 849–856, Aug. 2022, ISSN: 1573-7357.

[13] Collaboration, C.-P. et al. (including **R. G. Freundt**), "CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope", *The Astrophysical Journal Supplement Series*, vol. 264, no. 1, p. 7, Dec. 2022, et al. including R. G. Freundt, ISSN: 0067-0049.

[14] T. Nikola, S. K. Choi, C. J. Duell, R. G. Freundt, Z. B. Huber, Y. Li, K. Malavalli, M. Niemack, K. M. Rossi, G. J. Stacey, E. M. Vavagiakis, B. Zou, N. F. Cothard, J. Austermann, J. D. Wheeler, J. Gao, M. R. Vissers, J. Hubmayr, J. Beall, and J. Ullom, "CCAT-prime: The epoch reionization spectrometer for primce-cam on FYST", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 247–262.

[15] B. Zou, S. K. Choi, N. F. Cothard, **R. Freundt**, Z. B. Huber, Y. Li, M. D. Niemack, T. Nikola, D. A. Riechers, K. M. Rossi, G. J. Stacey, and E. M. Vavagiakis, "CCAT-prime: The design and characterization of the silicon mirrors for the Fabry-Perot interferometer in the Epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 982–994.